

LIFE IN THE UNIVERSE

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THE ONLY KNOWN LIFE IN THE COSMOS is that on Earth. Life on Earth is so ubiquitous, however, and the universe so enormous, that many scientists think there is a very good chance that life

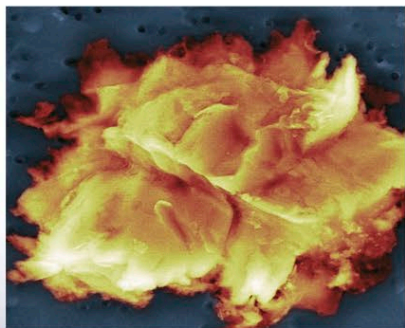
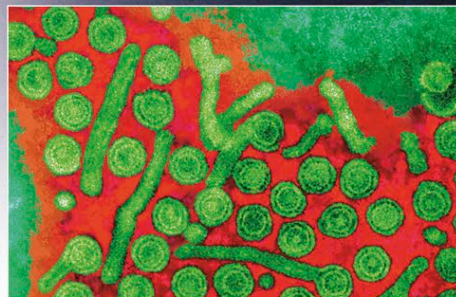
also exists elsewhere. Much depends on whether the development of life on Earth was a colossal fluke—the product of an extremely improbable series of events—or, as many believe, not so unexpected given what is suspected about primordial conditions on the planet.

LIVING ORGANISMS

What exactly constitutes a living organism? Human ideas on this are heavily reliant on the study of life on Earth, since scientists have no experience of the potential breadth of life beyond. Nonetheless, biologists have agreed on a few basic features that distinguish life from nonlife anywhere in the cosmos—as a bare minimum, a living entity must be able to replicate itself and, over time, to evolve. Beyond that, the definition of life is not universally agreed on. As an illustration, there is uncertainty about whether viruses are living. Although they self-replicate, viruses lack some characteristics that most biologists consider essential to life—notably, they do not exist as cells or possess their own biochemical machinery. It is also uncertain that other characteristics common to life on Earth, such as carbon chemistry or the use of liquid water, must inevitably be a feature of extraterrestrial life. Disagreements over such matters add complexity to discussions of the likelihood of life beyond Earth.

VIRUS PARTICLES

Viruses, such as this hepatitis virus, are on the border between living and nonliving matter. They self-replicate but can do so only by hijacking the metabolic machinery of animal, plant, or bacterial cells.



ORIGINS OF LIFE

Most scientists agree that the beginnings of life on Earth were linked to the accumulation of simple organic (carbon-containing) molecules in a “primordial soup” in Earth’s oceans not long after their formation. The molecules originated from reactions of chemicals in Earth’s

SUBZERO LIFE FORM

This so-far-unclassified life form was found living deep in the Antarctic ice sheet. Life can exist in a wider range of conditions than once thought.

atmosphere, stimulated by energy, perhaps from lightning. Within the soup, over millions of years the organic compounds reacted to form larger and more complex molecules, until a molecule appeared with the capacity to replicate itself. By its nature, this molecule—a rudimentary gene—became more common. Through mutations and the mechanism of natural selection, variants of this gene developed more sophisticated survival adaptations, eventually evolving into a bacterial-like cell—the precursor of all other life on Earth.

STROMATOLITES

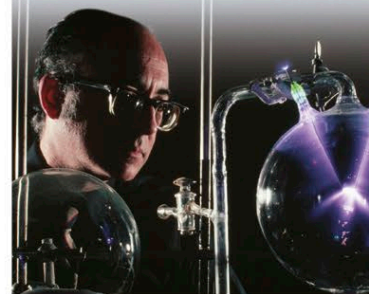
Some of the earliest remains of life are fossil stromatolites—mineral mounds built billions of years ago in shallow seas by cyanobacteria (blue-green algae). Stromatolites still grow on the Australian coast (left).

Many evolutionary biologists would say that the decisive event was the appearance of the self-replicator, after which living organisms would inevitably follow.

EXPLORING SPACE

RECREATING PRIMORDIAL EARTH

In 1953, American chemist Stanley Miller (1930–2007) recreated what he thought was Earth’s primordial atmosphere in a flask. He sent sparks, simulating lightning, into the gas mixture, which lacked oxygen. The result was many different amino acids—some of the basic building blocks of life.



STANLEY MILLER

Here, Stanley Miller recreates the experiment he first conducted as a graduate student. It showed that amino acids could have formed in Earth’s oxygen-free early atmosphere.